

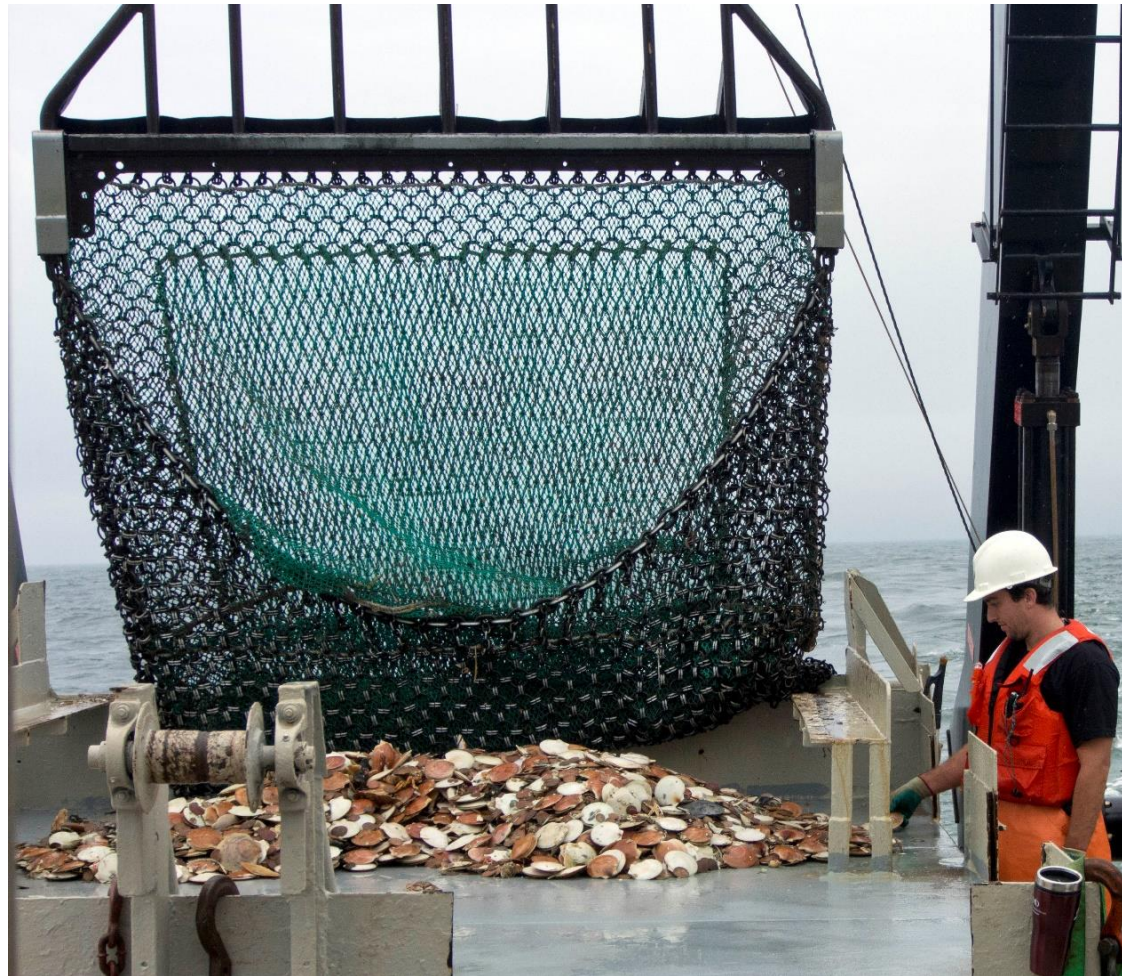
Term of Reference 3

Present the survey data used in the assessment (e.g., indices of relative or absolute abundance, recruitment, state surveys, age-length data, application of catchability and calibration studies, etc.) and provide a rationale for which data are used. Describe the spatial and temporal distribution of the data. Characterize the uncertainty in these sources of data.

Dredge Surveys

Conducted annually since 1979 with an 8-foot modified New Bedford scallop dredge with 51 mm rings and a 38 mm liner, random-stratified design

Tows are 15 minutes @ 3.8 knots



Sorting and measuring the catch



Survey Vessel History

NEFSC Dredge Surveys

Conducted on research vessels, mainly the R/V *Albatross IV* (1979-1989, 1993-2007), and the R/V *Hugh R. Sharp* 2008-2022

VIMS Dredge Surveys

Conducted on commercial vessels, towing a lined survey dredge and a commercial dredge

Used to supplement NEFSC dredge survey in some areas, 2005-2013, but has been used as primary dredge survey in Mid-Atlantic since 2014, and also selected areas on Georges Bank. Full Georges Bank survey in 2023.

Vessel Calibration

The *Albatross IV* and *Hugh Sharp* were both calibrated to a commercial fishing vessel, the *F/V Nordic Pride*.

The comparisons suggest no differences between the *Albatross IV* and *Nordic Pride*, but ~5% greater catch on the *Sharp*. Dredge sensors and videos indicate that this is because the dredge on the *Sharp* is covering about 6% more ground. This is accounted for by normalizing tows to a standard tow length (1 nm).



Dredge Calibration with Habcam

Based on about 150 paired dredge/Habcam tows, dredge efficiency was estimated as $q = 0.4$ on sand/soft bottoms, and $q = 0.27$ in hard bottom areas (South Channel and Closed Area II North), see Miller et al. 2019 CJFAS 76:847-855.



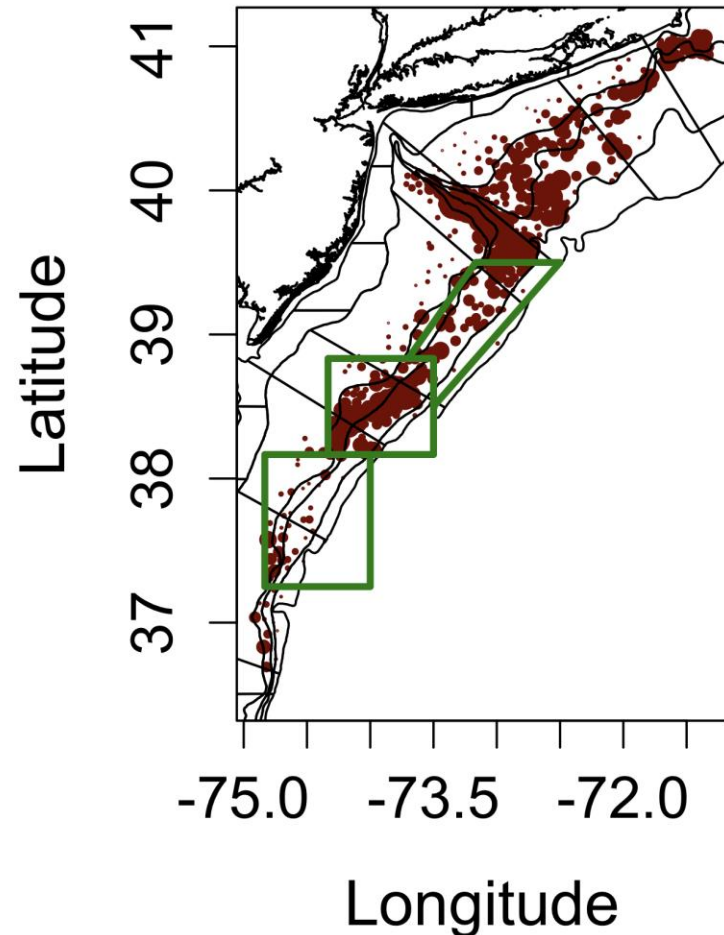
2022 Dredge Survey

Black lines are strata boundaries

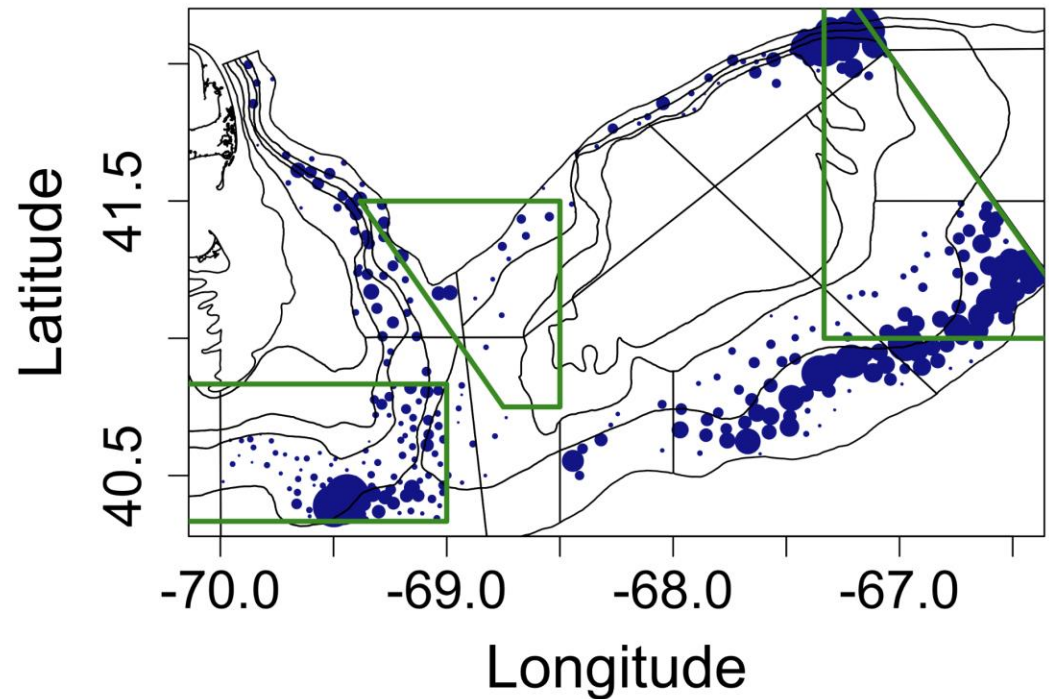
Green polygons are rotational areas

Area of dots proportional to catch weight

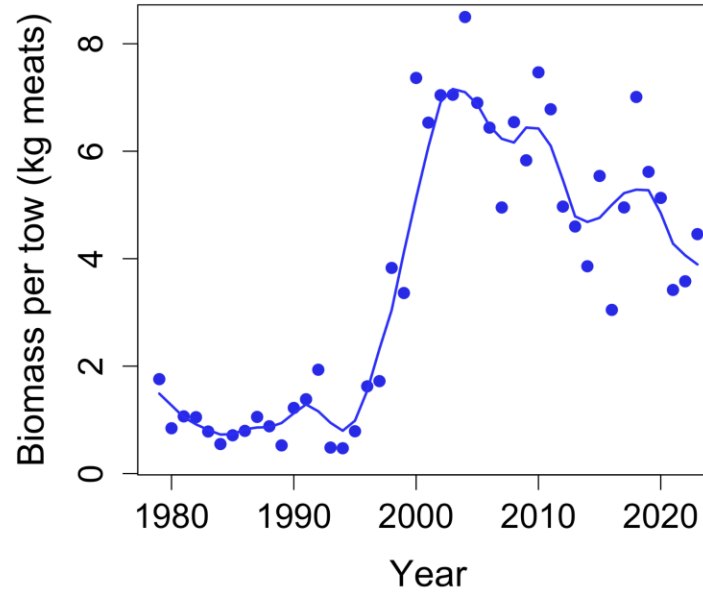
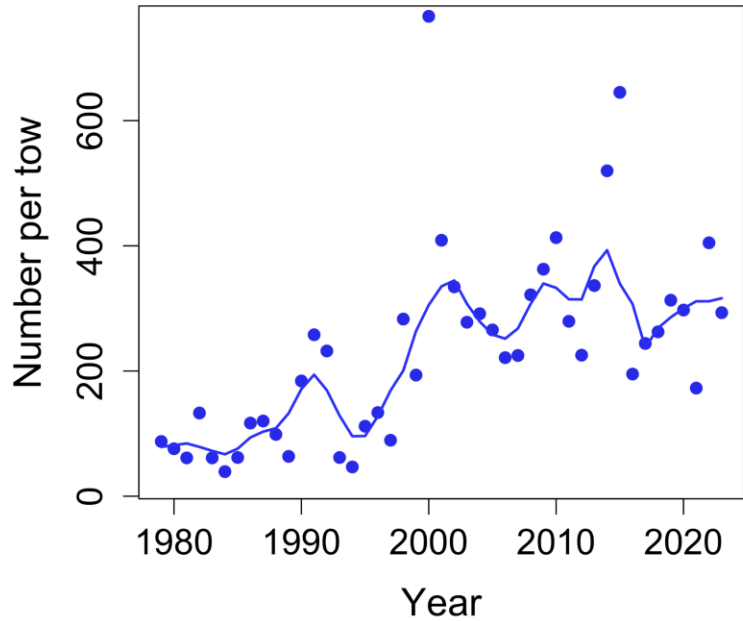
Mid-Atlantic



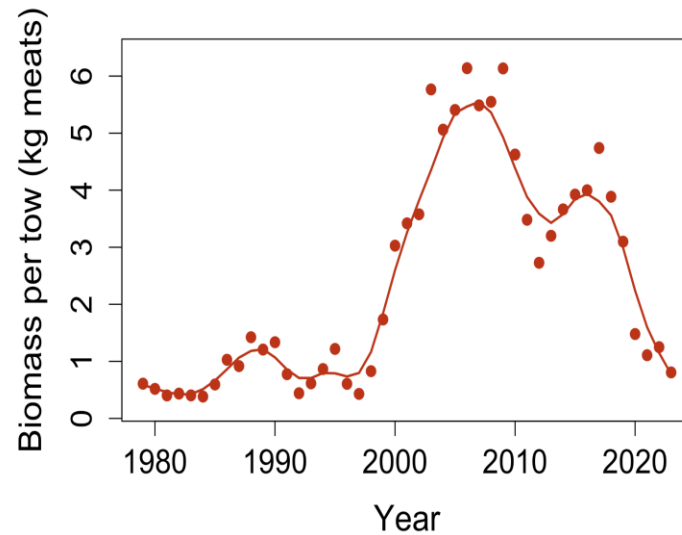
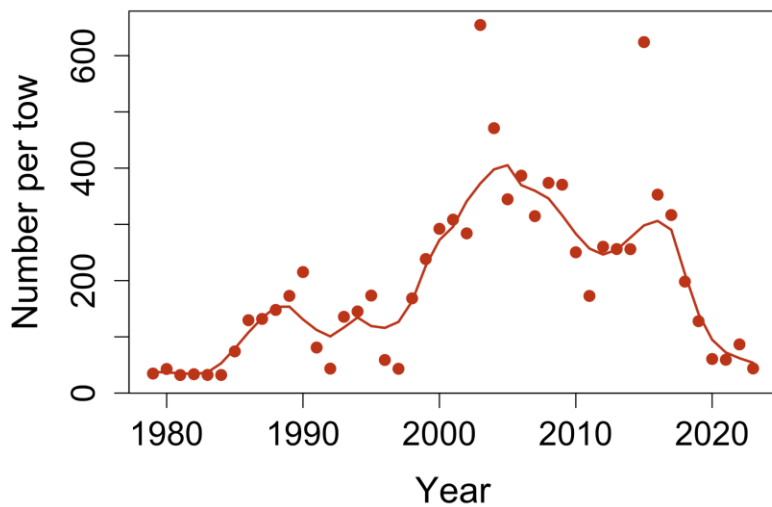
Georges Bank



Dredge Survey Time Series



Georges
Bank

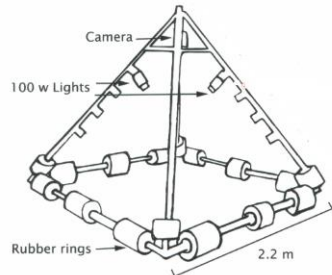


Mid-
Atlantic

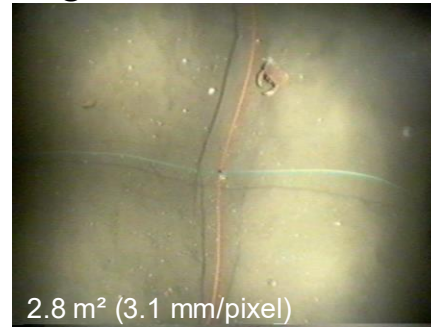
SMAST Drop Camera Survey

4 drops per station, stations 3 nm or less apart

1999



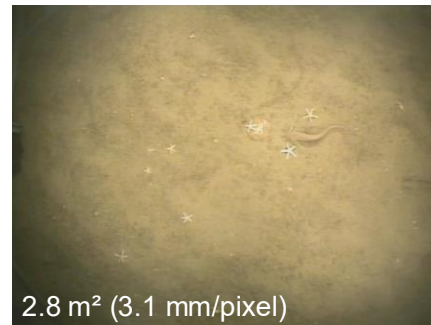
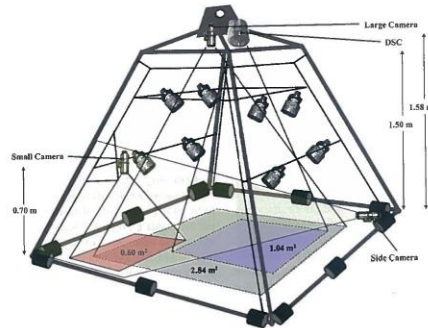
Large Camera



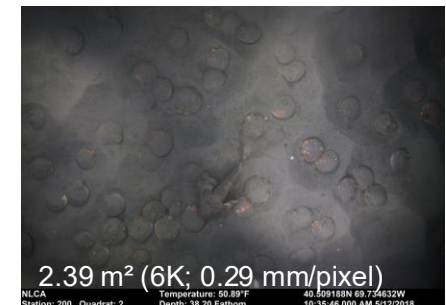
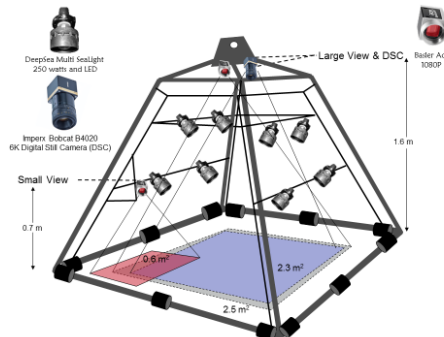
Digital Still Camera

N/
A

2009



2017

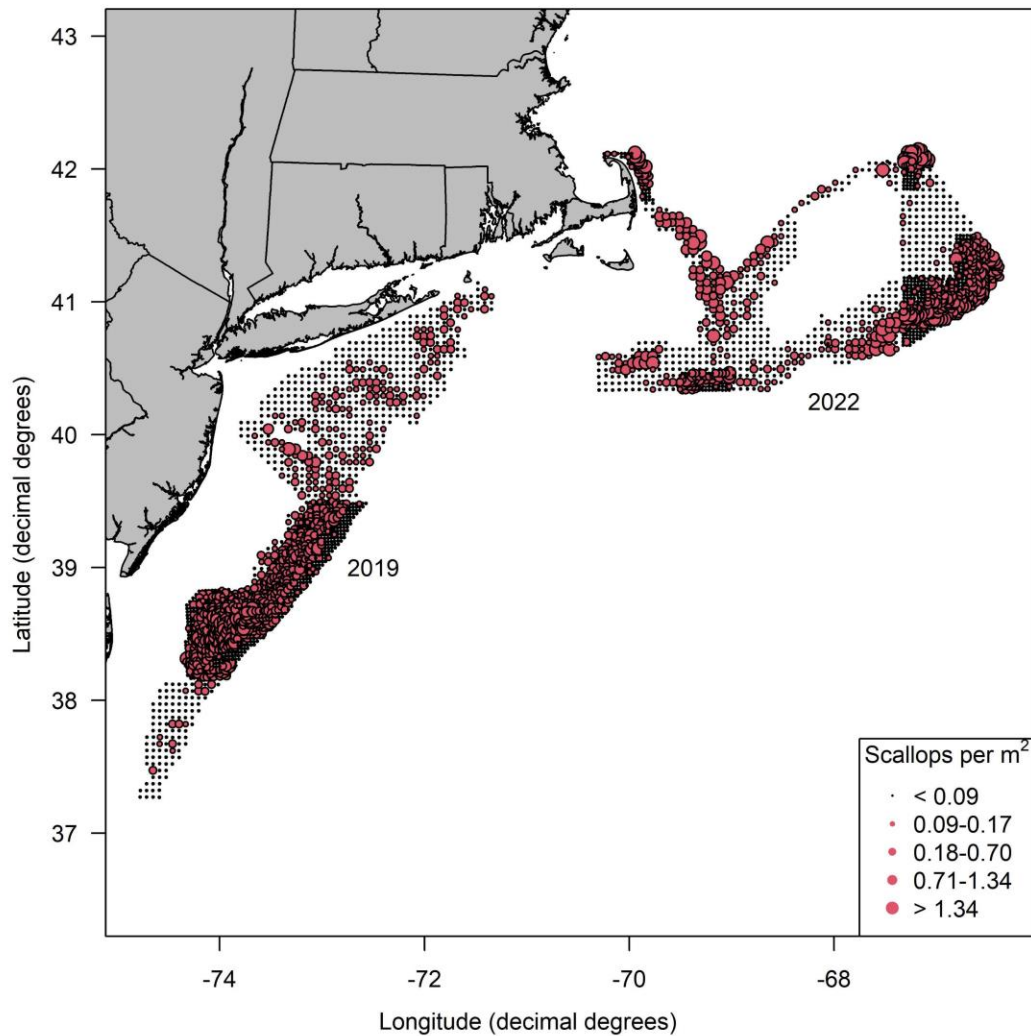


Camera and Field of View Specifications

Table 1. Time periods and resolution of cameras used on the drop camera survey. The large and small cameras are the same type of analog video camera (DeepSea® MultiSeaCam), but the large was mounted higher on the sampling pyramid resulting in a larger view area and lower resolution. The other three cameras all provided high resolution digital still images.

Camera	View area (m ²)	Vertical (mm pixel ⁻¹)	Horizontal (mm pixel ⁻¹)	Years used
Large	2.8	2.98	3.10	1999–2016
Small	0.6	1.35	1.43	2000–2016
Nikon	1.0	0.32	0.32	2007–2013
Kongsberg	1.7	0.41	0.41	2015–2017
Imperx	2.3	0.29	0.29	2017–present

Survey Years



Area	Years
Georges Bank	2003-2012, 2015, 2017-2023
Mid Atlantic	2003 – 2015, 2017, 2019
Gulf of Maine	2009 – 2011, 2013 – 2015, 2017, 2018, 2020 – 2023

Changes through time

- Cameras
- Grid resolutions
- Circa 2010 began accounting for ‘edge effect’ in images
- For the 2025 Research Track, GAMs were used to standardized the spatial footprint of the survey in each of Georges Bank and the Mid Atlantic

Large Camera Calibrations

“Large” video camera does not fully detect scallops less than 70-80 mm SH. “Small” video camera is closer to bottom and has smaller field of view, but can fully detect scallops down to ~40 mm

Selectivity of large camera estimated by comparison to the small camera (NEFSC 2007). Used large camera data with selectivity set by this comparison in the CASA model.

Measurement error in shell heights for large camera quantified in Jacobson et al. (2010) for scallops in test tank (SD of about 6 mm), but is larger in the field; it was estimated at 11.6 mm based on comparisons with the dredge survey.

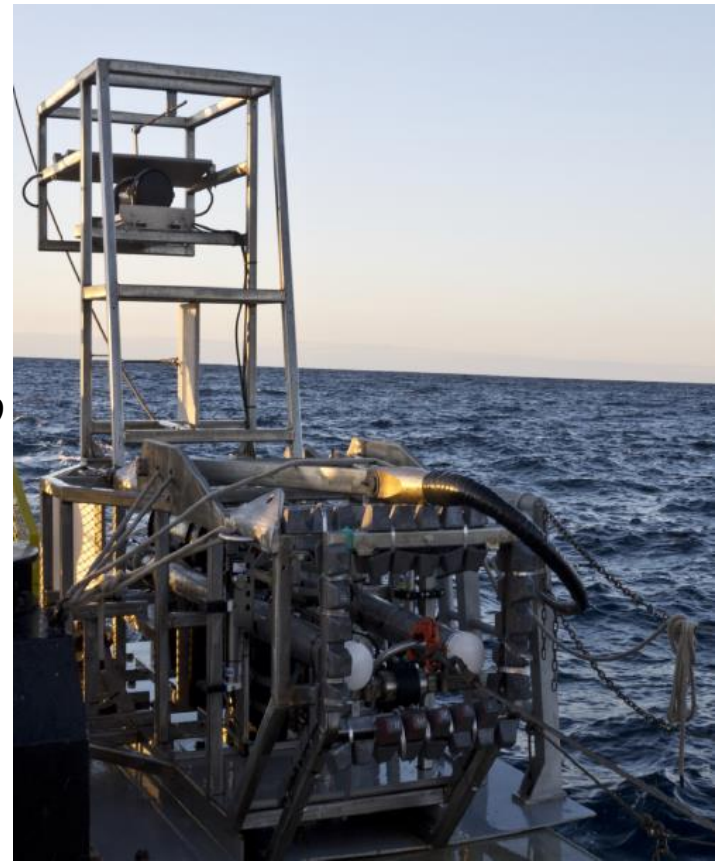
Habcam towed camera systems

Towed camera systems, taking 4-6 images/sec, towed at 5-6 knots at about 2 m above sea floor. Habcam v2 used a single camera, whereas Habcam v3, v4, v5 are all dual-camera stereo systems. Habcam v4 has been in use since 2012 on NOAA surveys. v2 (2005-2015) and v3 (2016-) have been deployed on commercial vessels and also on the *Sharp* in 2011 only.



Habcam v3
on the F/V
*Kathy
Marie*

Habcam v4
on the R/V
Hugh Sharp

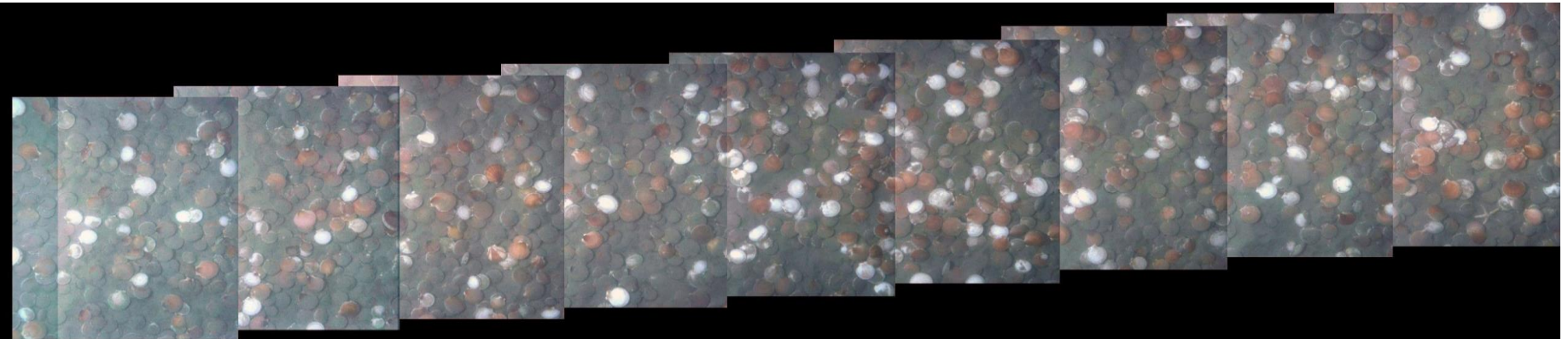


Habcam towed camera system

Between 1/50 to 1/100 images (about 1 per 20-40 m, or about 100,000 per survey) are manually annotated for scallops, fish and substrate. Developing automated annotators, which work quite well.

Main transects in the direction of the population gradient, which typically for scallops is perpendicular to the isobaths. Alternate long transects, which go out to the edge of the habitat, and short transects, which cover the “core” high density sections.

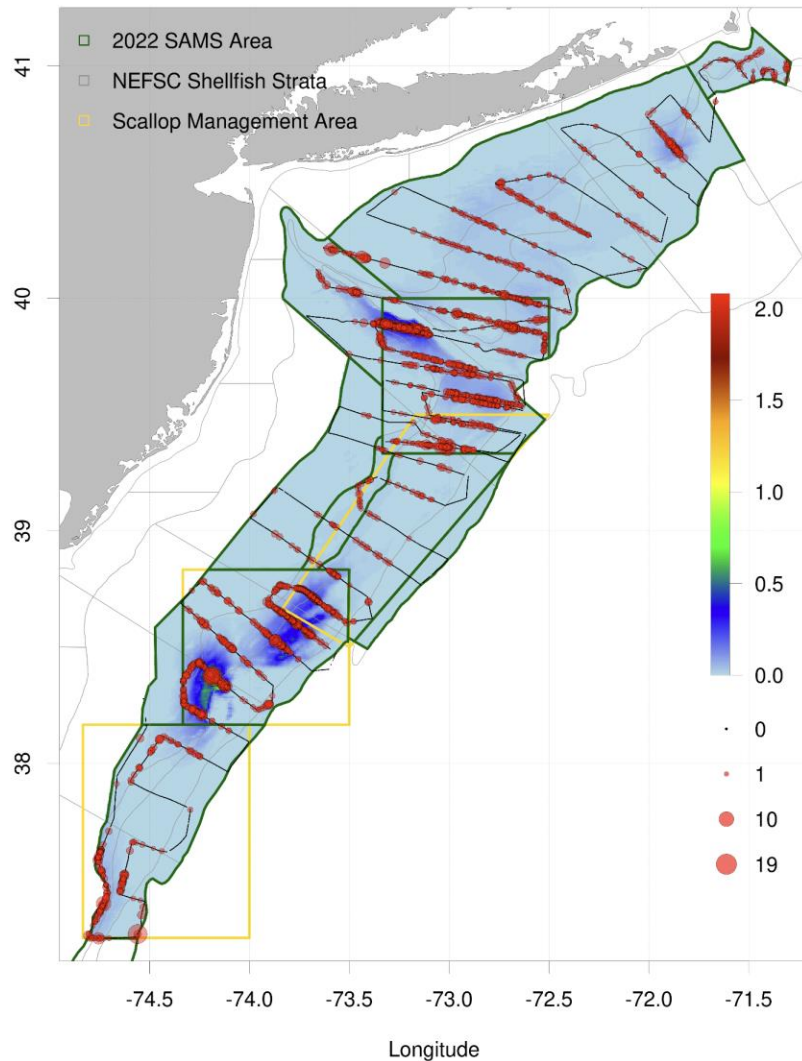
Data is interpolated using a form of regression kriging, where the data (aggregated into 750 m segments) is fit to a GAM, and then ordinary kriging is performed on the residuals (Chang et al. 2017)



2022 Habcam survey

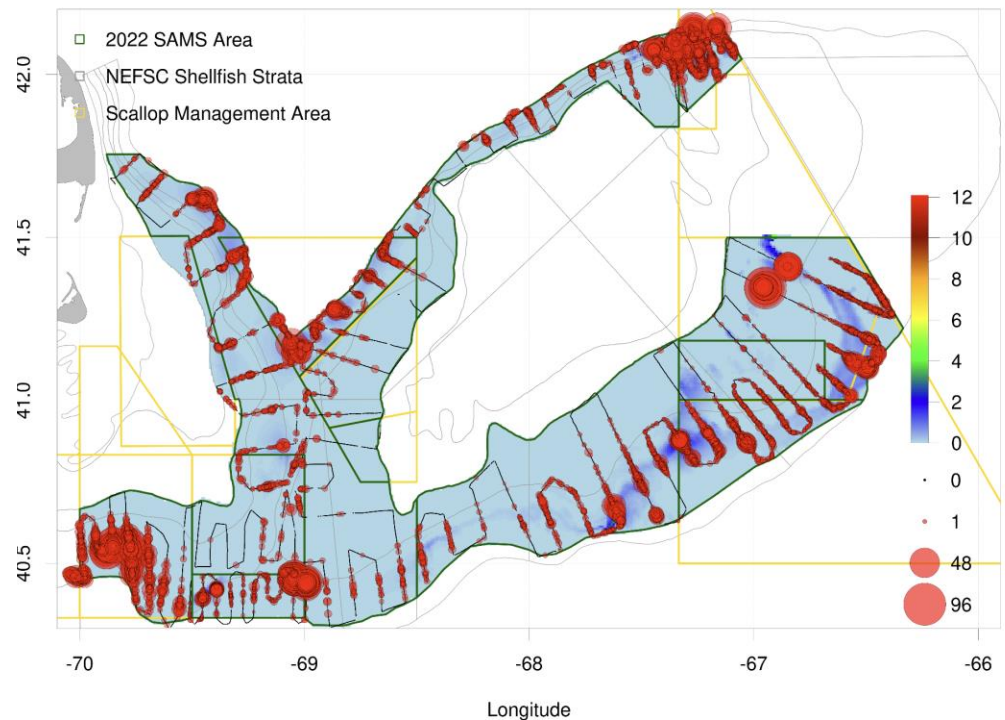
Mid-Atlantic

Prediction: # per m2 for shell height 40+ mm
Observation: # per m2



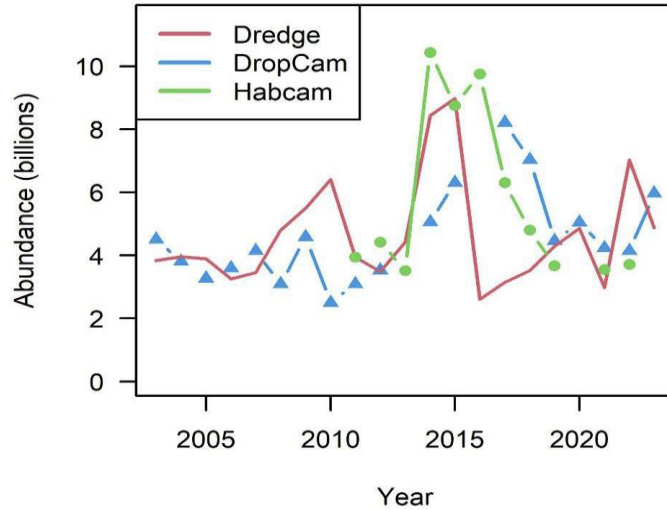
Georges Bank

Prediction: # per m2 for shell height 40+ mm
Observation: # per m2

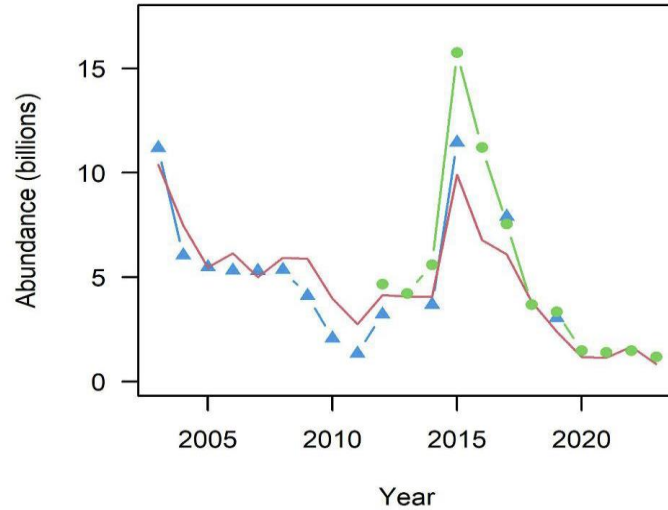


Survey Comparisons

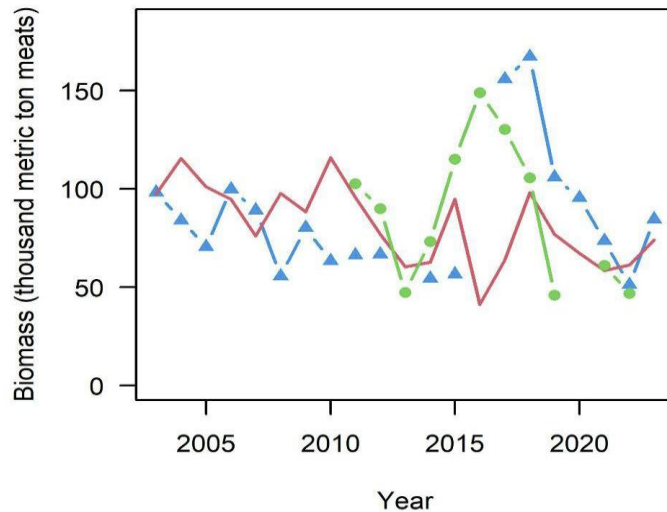
Georges Bank



Mid Atlantic



Georges Bank



Mid Atlantic

